Lecture 11: From Euler to Hamiltonian

When the requirement changes from covering connections to visiting locations

Euler

Cover every EDGE

→

Hamilton

Visit every VERTEX

Euler Graphs: Where Do They Matter?

The connections themselves must be serviced

Pipeline inspection robot — inspect every pipe segment and return to base.

Street sweeping / snowplow — every required street segment must be covered.

Power or communication cable inspection — every physical link must be checked.

PCB trace inspection — traverse all required electrical connections.

Garbage collection — service every required road segment.

Core requirement: cover every CONNECTION efficiently.

Now Change the Requirement

Suppose we no longer need to inspect every connection.

Instead, every LOCATION must be visited exactly once.

After visiting all required locations, we may need to return to the starting point.

Does such a route exist?

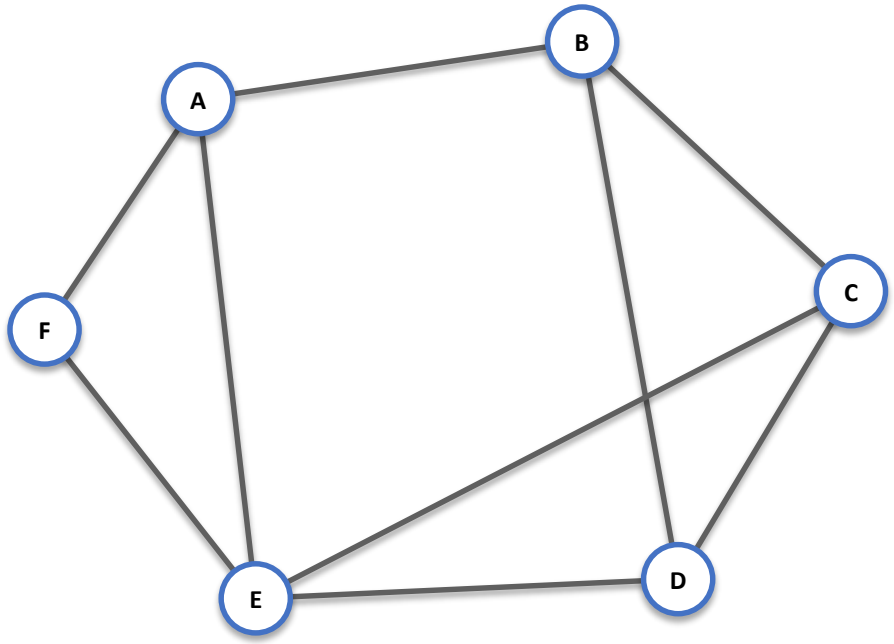
Practical Problem 1: PCB Drilling

A CNC machine must drill each required hole on a PCB.

Each drilling location must be visited once.

Allowed machine movements define which locations can follow one another.

The machine should return to its home position after completing the board.



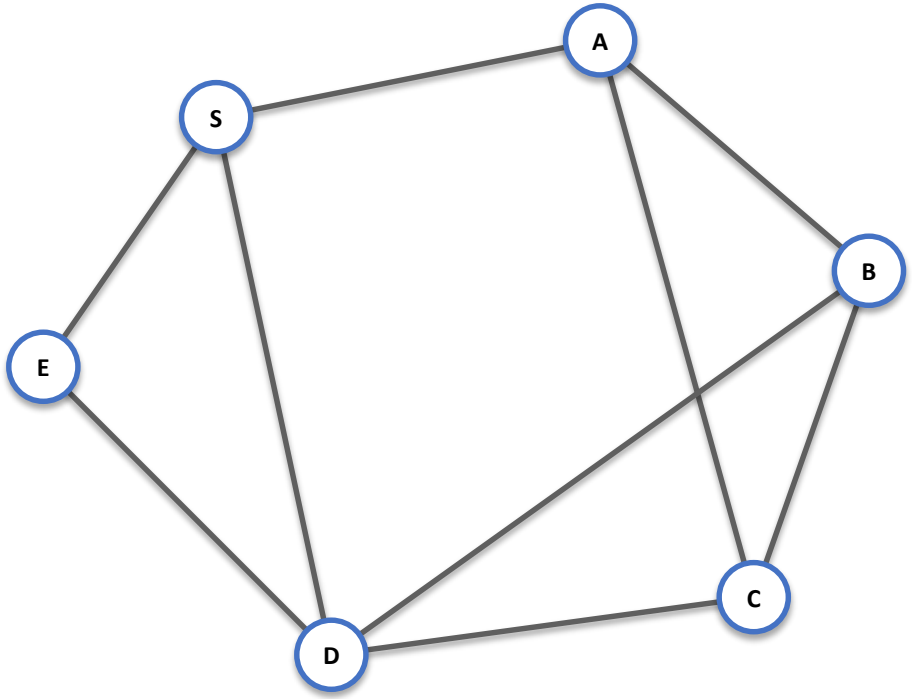
Can every drilling point be visited exactly once and the machine return home?

Practical Problem 2: Drone Inspection

A drone must visit every monitoring station exactly once.

Some direct movements are unavailable because of obstacles, restricted airspace, or range.

The drone must finally return to its base.



Feasibility first — not minimum distance yet.

The Common Structure

Visit every VERTEX exactly once

Hamiltonian Path: visits every vertex exactly once.

Hamiltonian Cycle: visits every vertex exactly once and returns to the starting vertex.

Euler vs Hamilton

	Euler	Hamilton
Target	Every edge	Every vertex
Typical question	Can every connection be covered?	Can every location be visited once?
Characterization	Simple degree condition	No comparable simple test
Computation	Hierholzer: O(V+E)	Hamiltonian Cycle: NP-complete

Structural Clues: When a Hamiltonian Cycle Is Impossible

Every vertex on a Hamiltonian cycle must have degree at least 2.

A graph with a Hamiltonian cycle cannot have a cut vertex.

A graph with a Hamiltonian cycle cannot have a bridge.

These are useful for ruling graphs out — but satisfying them does NOT guarantee a Hamiltonian cycle.

Dirac's Theorem

For a simple graph with $n \geq 3$:
$\delta(G) \geq n/2 \Rightarrow$ G is Hamiltonian

Meaning: if every vertex is connected to at least half of all vertices, a Hamiltonian cycle is guaranteed.

Important: if the condition fails, we cannot conclude that the graph is non-Hamiltonian.

Ore's Theorem

For every nonadjacent pair u, v:
$\deg(u) + \deg(v) \geq n \Rightarrow$ G is Hamiltonian

Ore can guarantee Hamiltonicity even when Dirac's minimum-degree condition is not met.

Both Dirac and Ore give sufficient conditions — not necessary conditions.

Practice 1 — Don't Misuse a Theorem

A simple graph has $n = 12$ and $\delta(G) = 5$.

Student claim: “The graph is not Hamiltonian because Dirac requires $\delta(G) \geq 6$.”

Is the conclusion valid? Justify in one sentence.

Think: What does failure of a sufficient condition actually tell you?

Practice 2 — Structural Reasoning

A connected graph has minimum degree 3.

It also contains a cut vertex.

Can the graph contain a Hamiltonian cycle?

Give a structural justification — do not search for a cycle.

Practice 3 — Dirac Fails, Ore Succeeds

A graph has 10 vertices and does not satisfy Dirac's condition.

For every pair of nonadjacent vertices u and v:

$\deg(u) + \deg(v) \geq 10$

What can you conclude? Which theorem supports your conclusion?

Practice 4 — Euler Does Not Mean Hamilton

A connected graph has an Euler circuit but contains a cut vertex.

A student claims: “Since we can traverse all edges in one closed tour, we can also visit every vertex exactly once in a closed tour.”

Evaluate the claim.

Practice 5 — Drone Network

A drone network contains 8 monitoring stations.

Every station has a direct connection to at least 4 other stations.

Without constructing the route, can you guarantee a closed tour that visits every station exactly once?

State the theorem and justify.

Takeaway

Euler asks whether every edge can be covered.

Hamilton asks whether every vertex can be visited exactly once.

Structural obstacles can prove that a Hamiltonian cycle is impossible.

Dirac and Ore can guarantee Hamiltonicity — but failure of their conditions does not prove non-Hamiltonicity.

Similar-looking traversal problems can have very different computational difficulty.